

Netting Out Exclusion: Deadwood, Under-Enrollment, and Voter Roll Quality in India

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Abstract

Democracies are judged by their turnout; yet before anyone can vote, they must be *registered*. We know little about who is on the voter list and who is not, especially outside the U.S. Standard accounts explain non-registration as a matter of costs. I argue that state capacity matters too: When a roll retains obsolete entries, this “deadwood” can mask – and sometimes produce – under-enrollment. Once rolls contain deadwood, aggregate measures of list quality – comparing population to enrollment totals – become uninformative, because they only capture *net* error. Recovering gross errors requires descending to the individual record. Comparing two full village censuses in India against the official electoral rolls, I show that both under-enrollment and deadwood co-exist, and that both errors fall disproportionately on women because of patri-local marriage norms and hyper-local registration. I trace the persistence of these errors to perverse incentives for street-level bureaucrats maintaining the rolls.

Keywords: voter registration, election administration, state capacity, gender, India

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1 Introduction

Voting is the most fundamental act of political participation in a democracy, as well as the main mechanism for holding officials accountable (Aldrich 1993; Przeworski 1999). Who votes – or doesn’t vote – matters for electoral outcomes, representation, and, more generally, the quality of democracy (Hill and Leighley 1992; Lijphart 1997; Verba et al. 2002; Griffin and Newman 2005). But who *can* vote is at least partly an administrative gate: only those who are on the voter list are actually allowed to cast a ballot.

A U.S.-centric literature has focused on voter registration requirements as a *cost* that particularly disadvantages the young, the poor, the mobile, and minorities, leading to differential under-enrollment among these groups (Rosenstone and Wolfinger 1978; Squire et al. 1987; Highton 1997; Ansolabehere et al. 2012). Yet maintaining a high-quality voter list is at least partly a state-capacity outcome (Shino et al. 2020; Morse 2023). After all, the voter list is the state rendering its citizens legible for the most consequential form of mass participation (Scott 1998; Lee and Zhang 2017; Brambor et al. 2020). In the absence of vital registers to underwrite automatic voter registration, most democracies rely on a mix of citizen initiative and state-sponsored enumeration efforts – from the Motor Voter Act in the U.S. (Highton and Wolfinger 1998), to localized enrollment camps in Kenya (Harris et al. 2021), to government-led canvassing drives in India (Quraishi 2019) – to create and maintain electoral rolls.

High-quality voter lists should be both *complete*, i.e., include all eligible citizens, and *accurate*, that is, contain no faulty entries or “deadwood” (James and Garnett 2024). The focus on registration requirements as *costs* can explain incomplete lists. A state capacity lens, on the other hand, can explain both errors of omission as well as errors of inclusion: a state that cannot fully see its population both fails to enroll people who belong on the list and fails to remove those who do not (such as the dead, migrants, or duplicates).

The quality of voter lists is conventionally judged – by scholars, election authorities, and policy-makers – by comparing roll totals against aggregate population numbers. Yet

this exercise only measures *net error* and is mathematically incapable of detecting *gross errors* (Sekar and Deming 1949), that is, it cannot tell us anything about the total number of exclusions and of erroneous inclusions/faulty entries. The consequence is potential disenfranchisement: where deadwood on the voter list offsets omissions, a voter list can look healthy while in fact being badly composed.

I demonstrate this using the case of India, the world’s largest democracy. I rely on unique data that enumerates the entire adult population of two villages and annotates the official voter lists for both locations, allowing me to recover gross errors in voter registration. I find that both villages simultaneously exhibit considerable under-enrollment of eligible citizens as well as deadwood on the voter lists. A closer look reveals that women and the young are the most likely to be unregistered; and female names on the voter list are more likely to be obsolete. Yet aggregate benchmarking is unable to detect these problems because deadwood on the voter lists masks under-enrollment. I attribute the double burden on women – in under-enrollment and deadwood – to gendered marriage norms that require women, but not men, to move once they marry.

I posit that the mechanism allowing deadwood to accumulate on Indian voter rolls is rooted in perverse bureaucratic incentives: because deletions from the voter list are more time-intensive and riskier than additions, and because the Election Commission of India (ECI) relies on *aggregate measures* of list quality that compare elector counts and gender balance on the lists against census estimates, frontline bureaucrats in charge of maintaining the list have incentives to skip deletions and cap additions. By not removing deadwood and simultaneously not enrolling newly eligible citizens, bureaucrats have a low-effort way of meeting aggregate benchmarks that is essentially undetectable using current monitoring strategies.

This article makes several contributions. First, it reframes the quality of voter rolls as a problem of coverage error rather than of completeness alone, distinguishing omissions from erroneous inclusions. Drawing on both the literature on voter registration and scholarship

on state capacity, I argue that registration costs can explain errors in one direction (omission) while limited information capacity generates errors in both. Second, and following from this, I show that standard aggregate benchmarks – comparing enrollment totals to population estimates – are structurally unable to detect gross errors once the voter list contains deadwood, because omissions and erroneous inclusions offset in the aggregate. Third, using novel record-matched data, I show that these errors are not demographically neutral in India. Patri-local marriage norms interact with hyper-local voter registration to produce both under-enrollment among women and deadwood among female names on the roll, which offset within the female electorate.

2 Framework

A core question in political science centers on who turns out to vote and why (see, for example, Downs 1957; Riker and Ordeshook 1968; Wolfinger and Rosenstone 1980; Rosenstone and Hansen 1993; Blais 2000; Green and Gerber 2015). But who turns out is necessarily bound by – and often strongly predicted by – who is *registered* to vote (Rosenstone and Wolfinger 1978; Erikson 1981; Squire et al. 1987; Timponi 1998; Highton 2004; Holbein and Hillygus 2016). Voter registration, therefore, is where the universal franchise is realized – or undermined – with essential consequences for the quality of democracy (Dahl 1956; Piven and Cloward 1989; Lijphart 1997; Verba et al. 2002; Griffin and Newman 2005).

There are two distinct dimensions of list quality: completeness and accuracy (James and Garnett 2024). Problems with completeness mean that some eligible adults are excluded from the list; problems with accuracy mean that faulty entries are contained in the register. There are also two distinct *sources of errors* in the voter list, studied very unequally in the literature. A largely U.S.-centric scholarship has focused on the costs that registration imposes on eligible citizens and the differential under-enrollment resulting from these costs. But a second, much less studied source of errors is based on state capacity, specifically the

state’s information capacity. Importantly, while a cost-based framework can explain under-enrollment, a state-capacity lens can explain errors in both directions.

Cost-based sources of under-enrollment are well theorized. Because returns to voting – and, by extension, to registering – are generally low, even small costs will have large deterrent effects, at least to some voters (Rosenstone and Wolfinger 1978; Aldrich 1993; Highton 1997). Travel distance, registration deadlines, residency requirements, and documentation requirements have all been shown to affect registration rates (Fleury 1992; Highton and Wolfinger 1998; Brady and McNulty 2011). Among eligible citizens, the young, frequent movers, the poor, and racial minorities are less likely to be registered (Piven and Cloward 1989; Keyssar 2000; Hanmer 2009; Ansolabehere et al. 2012; Braconnier et al. 2017; Huber et al. 2021; Harris et al. 2021; Fraga et al. 2023), while the effect of education on enrollment seems less clear (Nagler 1991; Highton 1997; Ansolabehere and Hersh 2012). Registration costs, therefore, can – and often do – produce under-enrollment.

While cost can explain under-enrollment, limited state capacity – specifically, information capacity – can explain both under-enrollment and inaccuracy. In order to maintain complete and accurate voter lists, the state needs to be able to “read” its population (Scott 1998; Lee and Zhang 2017; Brambor et al. 2020), and the two dimensions of list quality require different types of information. Completeness is a question of “stocks”¹: who lives in any given location, and who among the residents is eligible – the observable state of the world at a given moment, such as the moment of census enumeration. This can be canvassed, based on residence. Accuracy, by contrast, is a problem of “flows”: who on the list has died, who has moved away, or who has become ineligible. These flows, or changes, are discrete events that must be detected (or reported) rather than observed. Citizens may report deaths, moves, and changes in status routinely – but they usually report them to the agency on which some benefit depends and not the electoral register. The information that accuracy

¹ In using the language of static “stocks” and dynamic “flows” to distinguish between the total composition or size of a population at a specific point in time and events that change the composition or size, I follow standard nomenclature in demography, which has long made this distinction.

requires is therefore generated elsewhere in the state, if it is generated at all, and the voter list’s accuracy turns on whether that information travels (Morse 2023).

Completeness, therefore, is a shared task while accuracy is necessarily a state task. As the cost framework suggests, eligible citizens may seek to be registered to vote but deterred by the costs of enrollment. At the same time, democratic states usually undertake some sorts of efforts to enable registration, given incomplete lists’ implications for the quality of democracy (Norris 2014; James and Garnett 2024). By contrast, accuracy is the squarely the state’s responsibility (Merivaki 2020; Morse 2023). Citizens have no incentive to keep voter lists free from deadwood; de-registration is not deterred by costs, it has no benefit for citizens to begin with. This is most obvious in the case of the deceased: a dead citizen cannot de-register, at any cost.

Crucially, registration costs produce error in one direction (under-enrollment), but limited state capacity produces error in two directions (both under-enrollment *and* erroneous inclusions). This is essential, as it shapes what we can see when we look at the voter list. Let O denote omissions and I erroneous inclusions; the *gross error* on a register then is $O + I$. But the standard instrument for assessing list quality – comparing the number of registered electors to an estimate of the eligible population (McDonald and Popkin 2001; James and Garnett 2024; Fowler and Southern 2025) – observes only the *net error*, namely $I - O$. Under the cost framework, which assumes that $I=0$ by construction, net and gross coincide and a shortfall is a count of eligible citizens missing from the list. But under the capacity framework, where $I>0$, omissions and erroneous inclusions enter the aggregate with opposite signs and cancel out (Sekar and Deming 1949). In aggregate, therefore, the net error is likely to hide the true extent of the gross errors. In other words, a voter list may look healthy in aggregate but be inflated by deadwood while simultaneously excluding eligible citizens.

For illustration purposes, imagine a location with 10 eligible citizens whose voter list contains 10 names. Yet a scenario where all 10 citizens are registered and the list contains no deadwood is observationally equivalent to one where no one is enrolled and all entries

are deadwood: in both cases, the aggregate population figure (10) equals the number of entries on the list (10). Unless we can pinpoint either actual registration rates among the eligible population or share of deadwood on the voter list, comparing the totals is uninformative. That is true even in a world where population totals and voter list totals do *not* match: A location with 10 eligible citizens whose voter list contains 9 names can still have a registration rate anywhere from 0 percent up to 90 percent, with at least 1 (but up to 9) eligible citizens unregistered. That is, the shortfall on the list only marks the lower bound of under-enrollment. Perhaps more importantly, even when the number of entries on the list *exceeds* the population, under-enrollment is still possible: If the voter list for a location with 10 residents contains 11 names, surely one entry is deadwood – but up to 11 names *can* be deadwood, allowing for zero percent registration.

Importantly, errors in both under-enrollment and deadwood fall unevenly across the population, which might further distort what we can learn from looking at list aggregates for certain groups. This is especially true for the highly mobile. Frequent movers are less likely to be on the list because in many countries – including the U.S., France, the U.K., and India – individuals have to update their registration when they move to a new jurisdiction. A recent move simply means the individual has had fewer opportunities to register at their new place of residence (Squire et al. 1987; Highton 2004; Ansolabehere et al. 2012; Braconnier et al. 2017). At the same time, the literature has identified migration as the main source of deadwood on U.S. voter lists (Ansolabehere and Hersh 2010; Stewart 2019; Pettigrew and Stewart 2024). In addition, residential mobility is usually higher among the young and racial minorities, meaning that both under-enrollment as well as deadwood among those groups is likely to be higher as well (Ansolabehere and Hersh 2012; Huber et al. 2021).

Recovering gross errors requires descending to the level of the individual record, as work that validates registrations case-by-case rather than in aggregate has done (Ansolabehere and Hersh 2010; Shino et al. 2020; Jackman and Spahn 2021). The most direct way to do this is to match the voter list, name by name, against an independent enumeration of the

population (Sekar and Deming 1949). This method can recover what the aggregate hides: the composition of the voter list’s failures, namely which residents are omitted, and which register entries constitute deadwood. In what follows, I apply this record-level strategy in the context of India.

3 The Context: Voter Registration in India

Voter registration and list maintenance for all national and state elections are the purview of the Election Commission of India (ECI).² Voter lists are decentralized in India: electoral rolls are prepared at the level of the polling station. A polling station is the physical location where all voters residing in a clearly defined catchment area will cast their vote on election day, usually a government school.³ As in the U.S., Indian voters have to re-register when they move. Because voter lists are prepared at the polling station level, individuals have to update their registration when they move from one polling station catchment area to another, even if both locations are within the same village.

While India is an active registration state, the ECI makes increasingly strong efforts to enroll voters Shani (2017); Singh and Roy (2019); Quraishi (2019). That includes regular revisions of the voter lists, during which new applicants are to be added, those who are deceased or have moved away are to be deleted, and any other necessary changes to the roll are to be made. Revisions are made at least once a year, and additional revisions are performed before elections (Election Commission of India 2023, chapter 9). In order to facilitate enrollment, the ECI has introduced “Booth-Level Officers” (BLOs) – low-level government employees in charge of maintaining the voter list. Each polling booth (and therefore, each voter list) has one Booth-Level Officer permanently assigned to it; ideally,

² By contrast, local-level elections, for village councils (*panchayats*) and municipal councils, are managed by the State Election Commission (SEC) for each state.

³ As per directions of the ECI, any one polling station should not have more than 1,500 voters registered in its electoral roll, although the limit under the law is 2,000 voters (Election Commission of India 2023, p. 51). The ECI also aims to locate polling stations in a manner that does not require any voter to walk farther than 2 kilometers to the locations.

the BLO resides herself in the catchment area of that polling station (Election Commission of India 2023, p. 30). The BLO is expected to make regular field visits to the area during which she conducts door-to-door canvasses, assists eligible adults in filling out the necessary paperwork to get on the rolls, and updates information on those who moved away or passed away.

Importantly, the ECI’s monitoring mechanisms rely entirely on cross-checking aggregate data. To gauge the health of the electoral rolls, officials use “electoral data in comparison with population data to check the gaps in electors population ratio, gender ratio and enrolment in youths, women and other deprived sections of society” (Election Commission of India 2023, p. 6). Essentially, election officers apply the decadal growth rate to the most recent census data (2011) for a polling station and then compare aggregate metrics in the voter list and the population projection, including overall number of registered individuals, the ratio of female to male entries, and the number of newly added young adults (p. 54). Any aggregate abnormalities trigger mandatory audits by supervisors, including “households with more than 10 electors; abnormal gender ratio, and the first 20 polling stations with highest number of additions or deletions” (p. 92). Higher-ups are also required to verify personally “[d]eleitions in polling stations where the number of deletions exceed 2% of the total electors in the voters’ list of the polling stations” (p. 93).

Voter list quality in India is hard to gauge. Many scholars and analysts rely on the very aggregate comparisons of population figures and registration numbers I argue are uninformative for extracting true registration and deadwood rates (see, for example Retnakumar 2009; Shariff and Saifullah 2018; Roy and Sopariwala 2019; Retnakumar and Hari Kurup 2021; Das 2024). A limited number of studies have tried to evaluate the *completeness* of Indian voter lists, with varying results. One study of 13 polling stations across 4 North Indian states found that 91 percent of households had at least one family member enrolled in the list, but did not look at individual enrollment (Joshi et al. 2023). A 2015 NGO report from Delhi found that only 51 percent of respondents were correctly registered (Pyle and

Nair 2015, p. 48). In addition, registration rates might be inequitable across groups, with Muslims and migrants less likely to be enrolled (Kumar and Banerjee 2017; Verma et al. 2019; Gaikwad and Nellis 2021b; Das 2024). There is little data on the *accuracy* of electoral rolls in India, either. In Delhi, almost one third of 3,000 entries sampled from the voter list in 2015 were obsolete Pyle and Nair (2015). Migration is a major driver of deadwood on the voter lists. One study found that 15 percent of names sampled from the electoral rolls in a North Indian district belonged to migrants who had permanently moved away in 2009 (Verma et al. 2019).

4 The Data

To empirically demonstrate the argument developed above – that aggregate benchmarks can suggest a voter list is healthy while hiding large, offsetting gross errors – I draw on data from two villages in the state of Uttar Pradesh in North India. Note that I do not claim that the two villages studied here are representative of all of India, nor that the error rates there can be extrapolated to the rest of the country. Instead, they serve as proof of concept: standard aggregate metrics of voter registration, even at the relatively fine-grained level of a single polling station, *can* hide large errors.

4.1 Setting

The study was conducted in Uttar Pradesh, India’s most populous state. Home to an estimated 250 million people, the state of Uttar Pradesh has the largest number of registered voters in the country and sends more members to parliament, called *Lok Sabha* in India, than any other state (80 of 543). It is generally treated as a bellwether for national elections.

The two study villages are located in Bijnor and Bahraich districts. I selected these districts because a companion voter survey had been fielded there, which relied on the districts’ structural differences: Bijnor is relatively better developed, while Bahraich is one

of the least developed districts in India (NITI Aayog 2021). In 2011, the year of the latest government census, about 25 percent of Bijnor’s 3.7 million residents were classified as urban. Average literacy stood at 68.5 percent, and almost 43 percent of households used electricity as their primary source of lighting. By contrast, only 8 percent of Bahraich’s 3.5 million inhabitants were classified as urban. The average literacy rate was just above 49 percent, and only 14.4 percent of household relied mainly on electricity. Importantly, while Uttar Pradesh in general exhibits high male labor migration rates, Bijnor’s male outmigration is generally higher than Bahraich’s (Tumbe 2015, 2018).

Within each district, I chose one village through a multi-stage randomization procedure, first drawing one Assembly Constituency at random from among those covered by the companion survey, and then randomly selecting one polling station within that AC. At each site, I then conducted a full census of individuals living in the catchment area⁴ and annotated the entire voter list. Data was collected in March 2022, just after the 2022 Uttar Pradesh state elections had concluded.

4.2 Village Censuses

Enumerators approached every household contained within the catchment area and interviewed at least one respondent from within that household. If no one was home at any particular house, enumerators tried again a few days later, or alternatively collected contact information for the residents from neighbors or the *pradhan* (village council president). Families that stay away from the village for extended periods of time but still maintain a homestead in the location were interviewed either at their second residence in a nearby town or by phone.

Enumerators collected information on the household, including religion, caste, and income

⁴ The Village in Bijnor district actually had two polling stations within the administrative village boundaries, and therefore two separate electoral rolls. I enumerated the population in the catchment area of one of those polling stations, namely the local primary school (and therefore the northern part of the village). For ease of notation, I will call the catchment area enumerated Village A, instead of “the northern part of Village A.”

category. They also created a roster of all adults aged 18 years or older who usually live in the household⁵ and collected basic demographic information such as gender, age, marital status, and education. For each listed adult, investigators then validated whether the individual was registered to vote by checking the person’s information against the official electoral roll. Enumerators also noted whether the person was first registered before or after marriage, and roughly how many years ago they first enrolled.

4.3 Voter List Annotation

Voter lists for the two villages were downloaded from the Chief Electoral Officer’s website in March 2022. In each location, enumerators collected information on all entries contained in the electoral roll. Indian voter lists contain the voter’s name; their husband’s or father’s name; their age; their gender; a house(hold) number;⁶ and their voter ID number. For individuals who were not encountered during the village census, we tried collecting information from family members; members of the village council; and/or notables within the village who are most likely to have information about others, including the village teacher and health worker.

Entries on the voter list were then classified into one of three categories, namely correct entries, deadwood, and undetermined. These differ not only in kind but in how confidently they can be classified as well how contestable an entry’s status is. First, I marked those who were enumerated as part of the census as *correct* entries. This classification is both certain and uncontestable: names in this group were verified against the census and found to be correct, and there is no ground for the registered voter (or anyone else) to dispute their inclusion in the list.

The remaining entries are categorized as either deadwood or undetermined, based on the

⁵ India’s residence rule for voter registration merely requires that a person be “ordinarily resident” in a place; temporary absence does not invalidate ordinary residence. I adopted a similar wording in the survey questionnaire.

⁶ The “house(hold) number” does not represent a physical address; but a shared household number often-times corresponds to some type of (distant) familial relation.

amount of information I could gather. *Deadwood* comprises entries for which I have enough information to confirm that they do *not* correspond to a current resident; this categorization is therefore certain. I further distinguish four types of deadwood, namely a) clerical errors, b) deceased individuals, c) marriage migrants and c) other migrants. First, clerical errors usually took the form of duplicate entries. Second, individuals that were identified by members of their family as having died were marked as such. Third, marriage migrants were those entries that a) were identified as migrants by family members, b) were not claimed as ordinarily resident by any interviewed household, c) did not maintain their own homestead in the village, and d) were explicitly identified as having migrated for marriage. Finally, “other migrants” include all those names that a) were identified as migrants by family members, b) were not claimed as ordinarily resident by any interviewed household, c) did not maintain their own homestead in the village, but were d) *not* marriage migrants. These were usually labor or economic migrants.

These four types differ not just in the source of the error, but importantly in whether their removal from the roll could reasonably be contested. For clerical errors, the deceased, and marriage migrants, it could not: a duplicate or a departed individual have no claim to remain on the list; nor does a marriage migrant who has formed a new household in a different locality. Accordingly, no person or party would dispute their deletion. Other migrants are different, not because their non-residence is less certain, but because their removal is disputable, based on a legal grey area in voter registration regulations that leaves open what “ordinary residence” means.⁷ They are therefore deadwood on the same evidentiary footing as the other three types, but the only type whose status a reasonable observer — or the migrant themselves — might contest.

Lastly, *undetermined* entries were those names on the voter list for which we were not able to track down any information. That means we were unable to locate family members, or gather information from village council members, the *pradhan*, or local notables such as

⁷ See Appendix ?? for more details

the village teacher or health worker about the individual.⁸

4.4 Sample

Village A is located in Nagina Assembly Constituency in Bijnor district; Village B is located in Nanpara Assembly Constituency in Bahraich District. Appendix Figure A.1 shows the location of the two study villages on a map.

Panel A of Appendix Table A.1 provides summary statistics for the population enumerated in both study sites. Village A was larger, with 1,175 adults across 319 distinct households, compared to Village B with 708 adults across 239 households. The two villages were comparable in gender composition (50 percent female in Village A compared to 47 percent in Village B) and age (40 years vs. 41 years), but differed in their ethnic composition. Village A was a Muslim majority village, with almost 95 percent of residents identifying as Muslim. All of the Muslim residents identified as belonging to the Other Backward Classes (OBCs), while all of the Hindu residents identified as belonging to the Scheduled Castes (SCs). By contrast, Village B was a Hindu majority village (94 percent). Close to 29 percent of respondents identified as belonging to the general caste category, more than 53 percent as OBCs, and almost 18 percent as Scheduled Castes. Average education and household income were much higher in Village A, with an average of 5.5 years of formal schooling and mean monthly income of almost 11,400 INR ($\approx$ \$136), compared to 4 years of education and a household income of a little over 6,300 INR ($\approx$ \$76) per month in Village B.

Panel B of Table A.1 shows summary statistics for the voter rolls of the two study sites. The voter list for Village A contained 1,072 entries, 48 percent of which were marked female. The register for Village B showed 825 names, of which 46 percent were marked female.

⁸ Potential sources for these “undetermined” entries include long-ago migration or death that are not remembered by informants; clerical errors; or outright fraud. However, in speaking with village notables, no indications or suspicions of fraud came up.

5 Analysis

Village A has 1,175 eligible citizens and 1,072 entries on the voter roll. Village B has 708 residents and 825 names on the list. A naive interpretation of this aggregate data would suggest that Village A suffers from modest under-enrollment to the tune of 103 adults, or 8.8 percent of the total population. Village B, in contrast, exhibits deadwood of 117 entries, or 14 percent of the voter list. Yet an individual-level record match refutes that interpretation.

5.1 Net Vs. Gross Errors

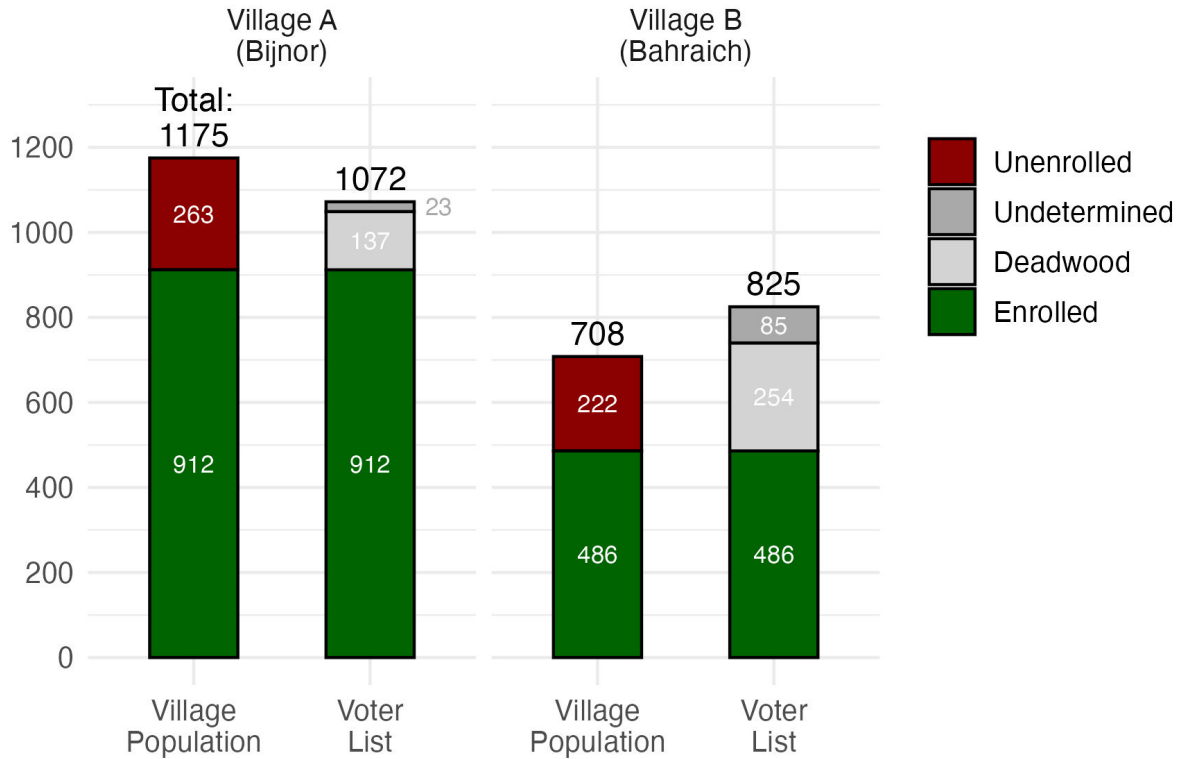
Figure 1 reveals that both villages suffer from under-enrollment *and* deadwood simultaneously – gross errors that cancel out and are therefore undetectable in the aggregate. The true scope of under-enrollment in Village A is 263 individuals, or 22.4 percent, not just the 8.8 percent suggested by the aggregate comparison. Because Village A’s voter list contains 12.8 percent of deadwood and 2.2 percent of undetermined entries, some of the under-enrollment is hidden in the aggregate. In other words, the net error underestimates true under-enrollment in the location by a factor of about 2.5.

Perhaps even more instructive is Village B. A total of 222 individuals – a staggering 31.4 percent of residents – are unregistered in a location where the voter roll *exceeds* the population. That is possible because the register contains deadwood to the tune of 30.8 percent of all names as well as 10.3 percent undetermined entries, not just the 14 percent excess suggested by the aggregate. An aggregate read of the data would have suggested the voter list is inflated, while in reality nearly one third of adults were missing from it.

5.2 Voter List Composition

The voter list consists of correct entries, deadwood, and undetermined entries.

Figure 1: Comparing Actual Village Population to Voter List



Correct Entries

The share of correct entries in the voter list was 85 percent for Village A and 58.9 percent is Village B. Since I have census information about these individuals, I will discuss details about this group in the population section below.

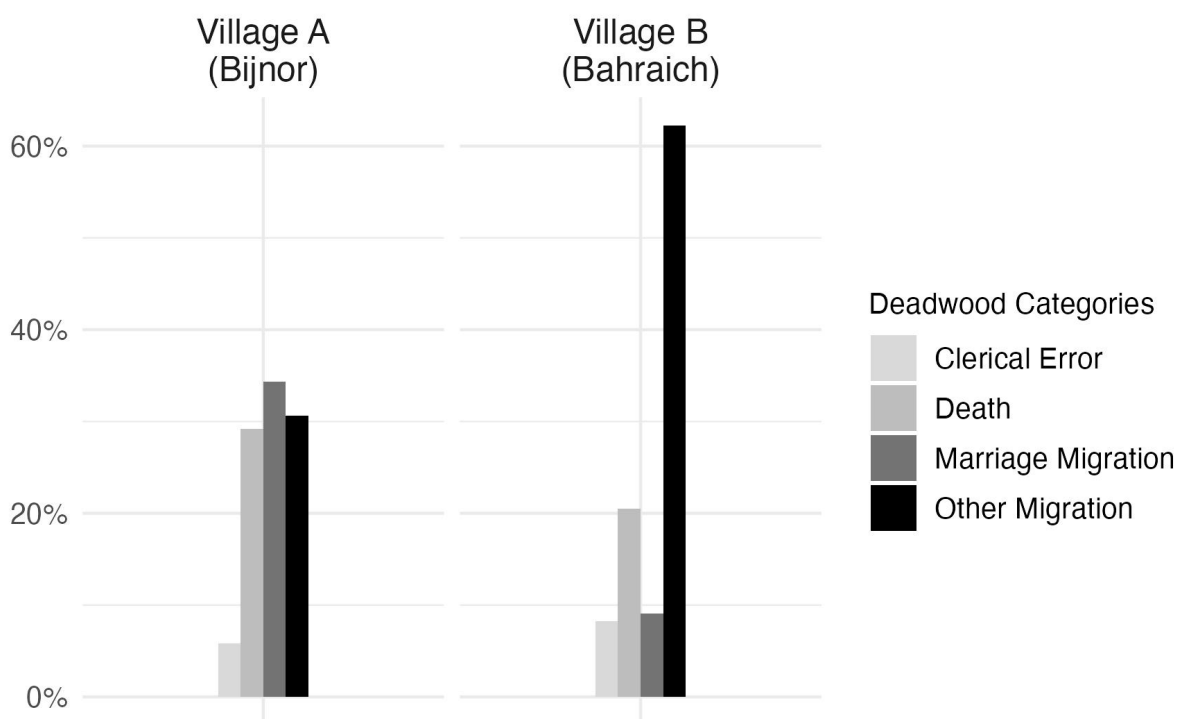
Deadwood

About 12.8 percent and 30.8 percent of entries, respectively, were classified as deadwood. I can disaggregate these entries into four deadwood types, namely clerical errors, deceased individuals, marriage migrants, and other migrants (see Figure 2). Recall that all deadwood is categorized with high certainty but varies in its contestability.

Only a small portion of faulty entries were clerical errors, namely 5.8 percent in Village A and 8.3 percent in Village B. All were duplicate entries. Entries representing the deceased

accounted for 29.2 percent and 20.5 percent, respectively. The share of entries classified as deceased far exceeds what we would expect given average mortality rates. The electoral rolls were last updated in January 2022. Even taking into account the high mortality rate caused by the delta-variant of the Covid-19 virus in India in 2021, average mortality rates in rural Uttar Pradesh in 2022 were around 8 per 1,000 per year (Office of the Registrar General of India 2026). Most of these entries, then, do not represent the recently departed, but residents who passed long ago but were never removed. Marriage migration accounted for 34.3 percent and 9.1 percent of deadwood entries, respectively. Finally, other migration accounted for a large share of entries classified as deadwood in both locations, namely 30.7 percent in Village A and 62.2 percent in Village B.

Figure 2: Decomposition of Deadwood Entries



I next look at the age and gender of entries classified as deadwood (see Appendix Table A.2). Those entries corresponding to deceased individuals unsurprisingly skew older: the

average age was 79 years in village A and close to 69 years in Village B. The fact that these entries are majority female is likely due to the fact that like in much of the rest of the world, women tend to live longer than men in rural India, meaning that among older residents, a higher share will be female.

As expected, all entries identified as marriage migrants were indeed female. The average age associated with those entries was also relatively young, at 26.4 and 29.5 years respectively. While that is considerably older than the average age at marriage for women, which was 19.3 years at the time of the 2011 census, this is likely an artifact of women who relocated for marriage remaining on the list for years after they move.

By contrast, the “other migration” category is majority male, with only 26 percent and 30 percent of entries in that group labeled female. It is also relatively young, especially in Village A, where the average age was 34 years, compared with 44 years in Village B. The fact that this group is largely male and of working age is in line with Tumbe (2018)’s description of semi-permanent labor migration. Furthermore, both the relatively lower share of women among these migrants and the relatively younger age in Village A suggest that Village A might see more male-only outmigration, during which only able-bodied men relocate in search of work while other family members stay behind. In contrast, the relatively higher share of women and the relatively higher age in Village B are consistent with more family migration, i.e., not just able-bodied men leaving but their wife and children joining them as well. This is further supported by the fact that more migrants on Village B’s voter roll share a household number. Recall that each entry on the voter list contains a household number, which does not necessarily correspond to co-residence but indicates at least *some* familial relationship. Conditional on having any migrant at all, households as indicated by this number have an average of 1.2 migrants in Village A, but 2.5 migrants in Village B, suggesting that more individuals belonging to the same family migrated in Village B, while usually only one individual from a family migrated in Village A.

Undetermined Entries

All entries that were neither matched to an actual resident nor confirmed as deadwood are classified as undetermined, a residual category that accounts for only 2.2 percent of entries in Village A but 10.3 percent in Village B. Given my data, it is impossible for me to resolve what these entries correspond to. However, the characteristics of these entries provide some preliminary clues. In Village A, these entries skew younger and majority male, consistent with labor migration. In contrast, the undetermined entries in Village B are majority female (77.6 percent) and older, on average (41.4 years). Looking only at male entries reveals that missing male entries are relatively young, at an average of 33.4 years. This is very comparable to the male undetermined entries in Village A, and close to the average age of male entries identified as labor migrants. The undetermined male entries in Village B might therefore potentially be economic migrants as well. The female entries tell a different story. The age distribution of undetermined female entries is bimodal, peaking around 30 and around 50 years. The early peak is compatible with marriage migration: entries that now show an average age of 30 might correspond to women who got married up to 10 years ago and were never removed from the list. In addition, the vast majority of entries older than 60 (87.5 percent) is female, raising the possibility that these undetermined entries might correspond to deceased individuals.

5.3 Who Is Enrolled – And Who Is Not

Having established what the electoral rolls contain, I now turn to the actual residents: who among the eligible population is registered, and who is left off? Before delving into the details, a word on what we can and cannot infer from two villages. I do not treat the associations reported here as representative estimates for India, and for most characteristics I make no generalizing claim at all. Whether characteristics like caste, religion, income, or education predict enrollment in my data is shaped by the particular composition of these two sites. Two findings are different in kind, however, and I make more general claims

about those. The overrepresentation of women and of the young among the unregistered does not rest on the idiosyncrasies of these two sites, but follows from two institutions that operate across India: patri-local marriage norms interacting with a hyper-local registration system. It is this mechanism, not the two-village estimate, that I theorize and defend below. I therefore report the full set of associations for completeness, but make substantive claims only for age and gender.

5.3.1 Comparing Enrolled and Unenrolled Residents

Table 1 provides a side-by-side comparison of registered residents and unregistered ones, by village. The table clearly shows that in both locations, the unenrolled skew female and younger. Women make up the minority of registered voters (46.3 percent in Village A and 42.4 percent in Village B), but the majority of unregistered adults (64.6 percent and 59 percent, respectively). The difference is substantively large and statistically significant ($p < .01$). Put as rates, women in these villages are substantially and statistically significantly less likely than men to be registered. Only 71.3 percent of female but 84 percent of male residents are registered in Village A; in Village B, enrollment is 61.1 percent for women and 75.5 percent for men ($p < .001$).

The unregistered are also significantly younger than the registered. In Village A, the average age for enrolled residents was 43.6 years, against 29.2 years for unenrolled ones. In Village B, the figures were 45.3 years and 32.4 years, respectively. In line with previous findings, enrollment rates increase with age (Ansolabehere et al. 2012). Only a small fraction of those aged 18 to 20 years old was registered to vote, namely 13.7 percent in Village A and 6.4 percent in Village (see Figure 3).⁹ By contrast, registration rates for 41-to-45-year-olds are 79.3 percent in Village A and 80 percent in Village B. In the age group over 50, registration reaches nearly 100 percent in Village A and close to 90 percent in Village B.

⁹ This holds when I exclude 18-year-olds who could, in theory, have reached voting age after the cut-off date. Even among 19- and 20-year-olds, registration rates were only 14.6 percent (Village A) and 7 percent (Village B).

Table 1: Comparing Registered and Unregistered Adults

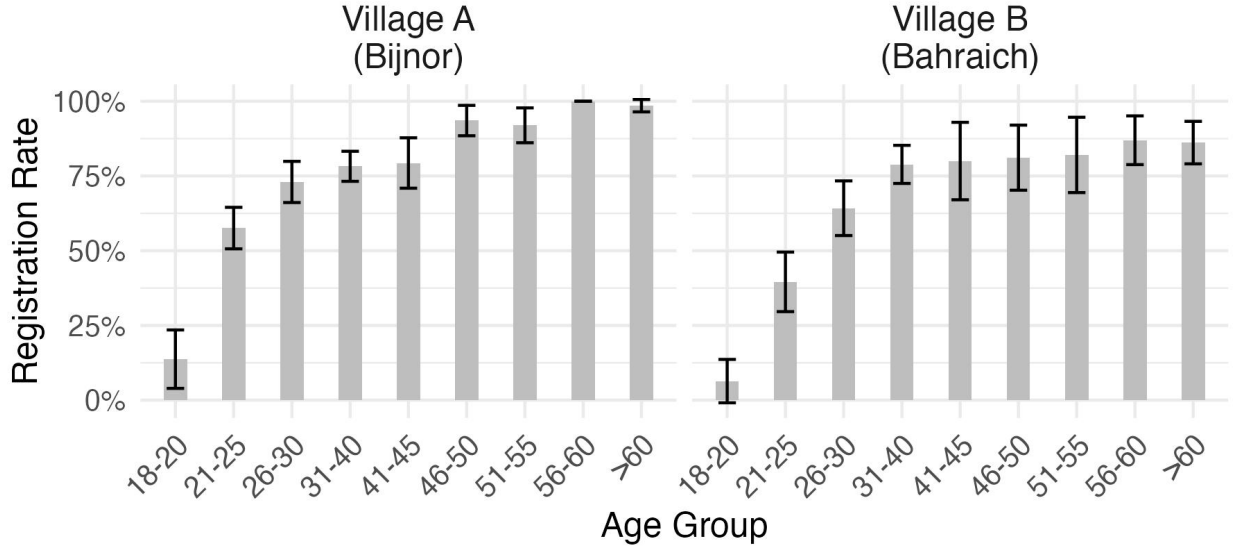
	Enrolled	Not Enrolled	Difference
Panel A. Village A (Bijnor)			
N	912	263	
% Female	46.3%	64.6%	-18.4***
Average Age	43.59	29.19	14.40***
Average Education	5.13	6.98	-1.85***
Average Household Income	13,308	12,966	342
% Hindu	4.9%	5.7%	-0.8
% Muslim	95.1%	94.3%	0.8
% General	0.0%	0.0%	0.0
% OBC	95.1%	94.3%	0.8
% SC	4.9%	5.7%	-0.8
Panel B. Village B (Bahraich)			
N	486	222	
% Female	42.4%	59.0%	-16.6***
Average Age	45.31	32.39	12.92***
Average Education	3.84	4.07	-0.23
Average Household Income	8,948	7,459	1,488
% Hindu	94.4%	92.8%	1.7
% Muslim	5.6%	7.2%	-1.7
% General	31.3%	23.4%	7.9*
% OBC	51.0%	58.6%	-7.5
% SC	17.7%	18.0%	-0.3

Notes: The Difference column reports enrolled minus unenrolled. Differences for binary variables are expressed in percentage points. Standard errors are clustered by household. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Education does not track registration status in expected ways, and the pattern is different between the two villages. In Village A, enrolled adults are *less* educated than unenrolled ones (5.1 years vs 7 years, $p < .01$). Yet the same does not hold for Village B, where the registered and unregistered have comparable levels of education, at 3.8 and 4 years, respectively.

Registered and unregistered individuals come from households with similar incomes. This is unsurprising, since under-enrollment is largely individual, not household-based, while income is measured for the whole household. In almost half of all households in the census, *all*

Figure 3: Registration Rate by Age



members were enrolled. Only a minority of households (5 percent) had *no* family members on the voter list. In the rest of households, some or even most members were registered, but at least one eligible adult was not on the list – at the same household income.

Perhaps unsurprisingly given the two villages relative ethnic homogeneity, enrolled and unenrolled adults are of similar religious and caste backgrounds within each village. Village A is almost entirely Muslim (95 percent), Village B is majority Hindu (94 percent), and in neither location do registered and unregistered adults differ meaningfully in their religious makeup. Their caste composition is likewise similar, with one notable exception: in Village B – which is the more diverse village along the caste dimension – registered voters are somewhat more likely to be general caste than unenrolled adults ($p < .1$), while the differences for OBCs and SCs are indistinguishable.

On the face of it, this might suggest that the minority under-enrollment documented elsewhere in India is absent here (Shariff and Saifullah 2018; Das 2024). I caution against drawing that conclusion. Each of my two villages is nearly homogeneous, so within-village comparisons are made among people who are overwhelmingly of the same religion to begin

with. My data cannot speak to whether religion is associated with enrollment in a setting where religion varies meaningfully, which is precisely the setting in which minority under-enrollment has been documented before. A similar logic holds for caste, which not only barely varies in Village A, but is entirely colinear with religion (all Muslims identify as OBC and all Hindus as SC). Only in Village B does caste vary enough to detect genuine differences – and it is the one setting in which a historically privileged group (general caste) makes up a slightly higher share of the registered than the unregistered.

5.3.2 Correlates of Enrollment

To more formally assess which characteristics are associated with enrollment, I estimate OLS models which regress an indicator for being registered on an individual’s gender, age, years of schooling, and household income. The specification includes a village indicator to capture baseline differences between the two study sites. Standard errors are clustered at the household level. The models do not include religion or caste, since both villages are religiously near-homogenous and caste is colinear with religion in Village A.¹⁰

Table 2 presents results. As the main specification (column 1) shows, all else equal women are about 11 percentage points *less* likely than men to be enrolled ($p < .001$).¹¹ Age is a strong predictor of enrollment as well: each additional year raises the probability of being on the list by more than 1 percentage point ($p < .001$). Education is significant ($p < .05$) in the pooled model, but the by-village results (Appendix Table A.3) show that this correlation is confined to Village A. Household income is unrelated to enrollment ($p = .36$). Overall, residents of Village B are about 8.7 percentage points less likely to be registered than adults in Village A, holding all other characteristics constant ($p < .001$).

The model with interaction effects (column 2) shows that the association between age and enrollment is *stronger* for women than for men ($p < .001$). This indicates that women’s

¹⁰ I present village-specific models which include caste for Village B, the more caste-diverse setting of the two in Appendix Table A.3.

¹¹ The coefficient on female in the interaction model (column 2) is the gap at zero age and education; at mean age and education, it is likewise about 11 percentage points.

Table 2: Correlates of Enrollment

	Main	With Interactions	No Village Control
Female	−0.114*** (0.019)	−0.280*** (0.071)	−0.109*** (0.019)
Age (years)	0.011*** (0.001)	0.010*** (0.001)	0.011*** (0.001)
Education (years)	0.007** (0.003)	0.005* (0.003)	0.008*** (0.003)
HH income (1,000 INR)	0.001 (0.001)	0.001 (0.001)	0.002* (0.001)
Female × Age		0.004*** (0.001)	
Female × Education		0.004 (0.004)	
Village B	−0.087*** (0.025)	−0.085*** (0.025)	
Num.Obs.	1883	1883	1883
R2	0.175	0.178	0.166

Robust standard errors clustered by household in parentheses. Religion and caste are excluded because they do not vary within village (see text). * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

probability of registration rises faster with age than men's. By contrast, the education gradient does not differ significantly by gender.

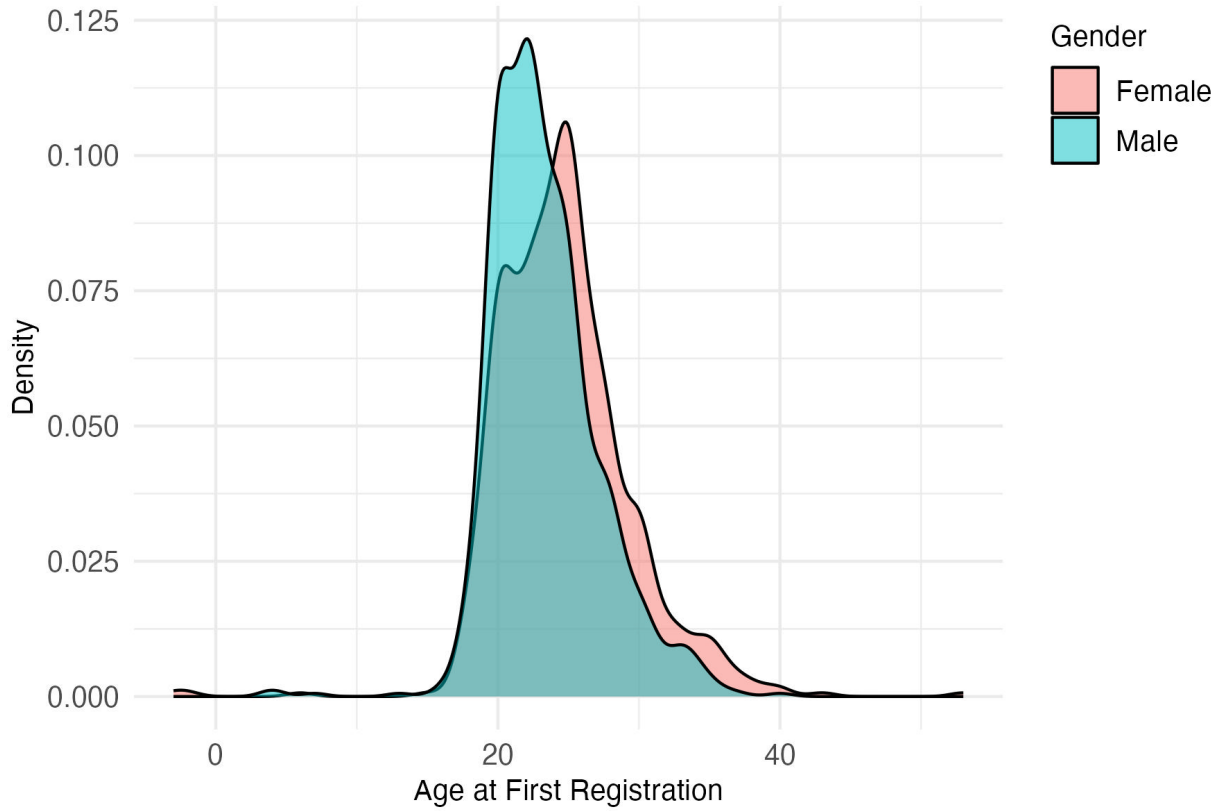
5.4 Marriage and the Timing of Women's Registration

So far, I have shown that women are significantly less likely than men to be registered and that women's probability of enrollment rises more steeply with age than men's. Here I argue that both findings reflect gendered registration *timing* in India, arising from the interplay of patri-local marriage norms and hyper-local voter registration. I make this argument inductively, by reading my data in light of well-documented, pan-Indian marriage norms, and showing that empirical patterns are consistent with this explanation. Data from my two villages serves as illustration but cannot establish the mechanism causally nor claim generalizability beyond these sites.

Looking at registered voters only, women were, on average, significantly older than men when they first enrolled (24.7 years vs 23.3 years, $p < .001$). Figure 4 visualizes approximate age at registration, calculated as current age minus years of registration. While first enrollment for both men and women peaked in their 20s, the peak is higher and earlier for men than women, and the right tail of the female distribution is much fatter. Women were also significantly more likely to only register *after* getting married, while the majority of men registered before marriage. Only 24.1 percent of female registered voters but 55.1 percent of males reported first getting on the list before marriage.

Both women's relatively higher age at registration and their higher likelihood of enrollment only after marriage can be explained by the interplay of patri-local village exogamy and the fact that registration is *polling-station-specific*, and therefore hyper-local, in India. Women across much of India are expected to marry outside their natal village and to join their husband's household upon marriage. While this norm is not universal, it is the most common kinship norm, particularly among Hindus and particularly in North India (Dube 1997; Jejeebhoy and Halli 2005; Desai and Andrist 2010; Brulé 2020). In fact, the 2011

Figure 4: Age at First Registration, by Gender



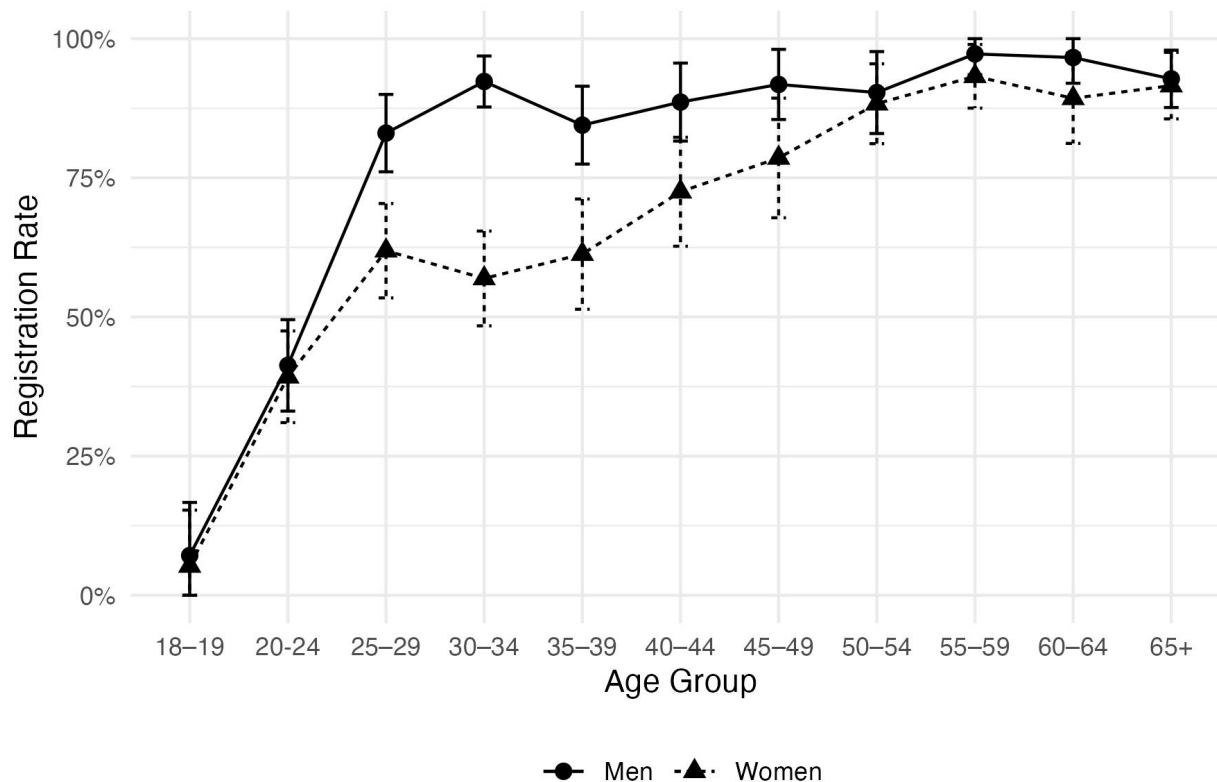
census counted a total of 211.2 million individuals who in the past had migrated for marriage, more than 97 percent of which were female (Census Table D-03).¹² Of those female marriage migrants, 72 percent (148.3 million) were rural-to-rural migrants; the majority of those female rural marriage migrants relocated within the same district (107.1 million, or 72.2 percent).

Because registration in India is tied to the polling station in whose catchment area a voter resides, even minor spatial moves – such as relocating to a neighboring village or an adjacent neighborhood in a town – will mandate re-registration. The U.S. literature has established that residential tenure increases the chance of being enrolled, and moving resets the clock because it requires re-registration (Squire et al. 1987; Ansolabehere et al. 2012). Patri-local village exogamy mandates a residential move for women, but not men: even

¹² In total, the census enumerated 455.8 million migrants, two-thirds of whom were women (68 percent).

at a gender-equal, constant probability of registering in any given year, then, men’s longer residential tenure means they had more opportunities to enroll compared to women. This will mechanically produce a gendered registration gap, concentrated in the marriage-migration age window.

Figure 5: Registration Rate by Age Group and Gender

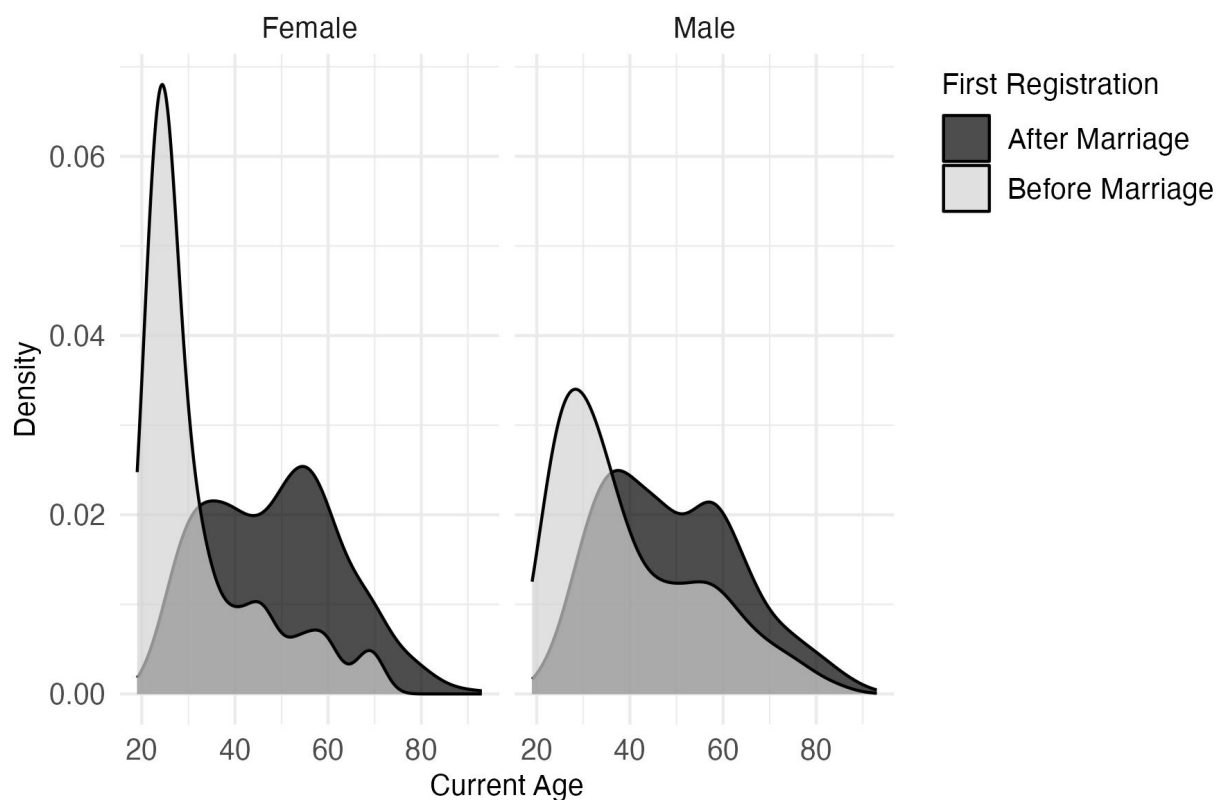


That is exactly what Figure 5 shows. In the youngest age groups, up until their early 20s, registration rates between men and women are essentially indistinguishable. But they diverge sharply after that, with male rates rising rapidly before plateauing around age 30, while women’s remain stagnant before catching up again in their 40s. However, there is essentially no gender registration gap among currently unmarried residents, with 54 percent of unmarried men and 55 percent of unmarried women on the list. The gap, instead, is concentrated among married individuals: 70 percent of married women but 90 percent of married men were enrolled (see Appendix Figure A.2). This is in line with a residential-

tenure story: even if women are on the list in their natal village, they have to re-enroll when they move for marriage, meaning that *after marriage*, women are less likely to be registered than men.

Yet a purely mechanical, tenure-driven explanation cannot account for differences between cohorts that I observe in my data. Specifically, while older women in my data were more likely to have first enrolled *after* marriage and to have been older at first registration, this pattern does not hold for younger women. Figure 6 shows that among registered voters, women who first enrolled *before marriage* were disproportionately young at the time of my study, with the distribution peaking around 25 years of age. By contrast, those who registered after marriage are spread more broadly across older cohorts, with substantial mass around middle age (40 to 60 years old). For men, the distributions are more similar, although the group that registered before marriage still skews younger.

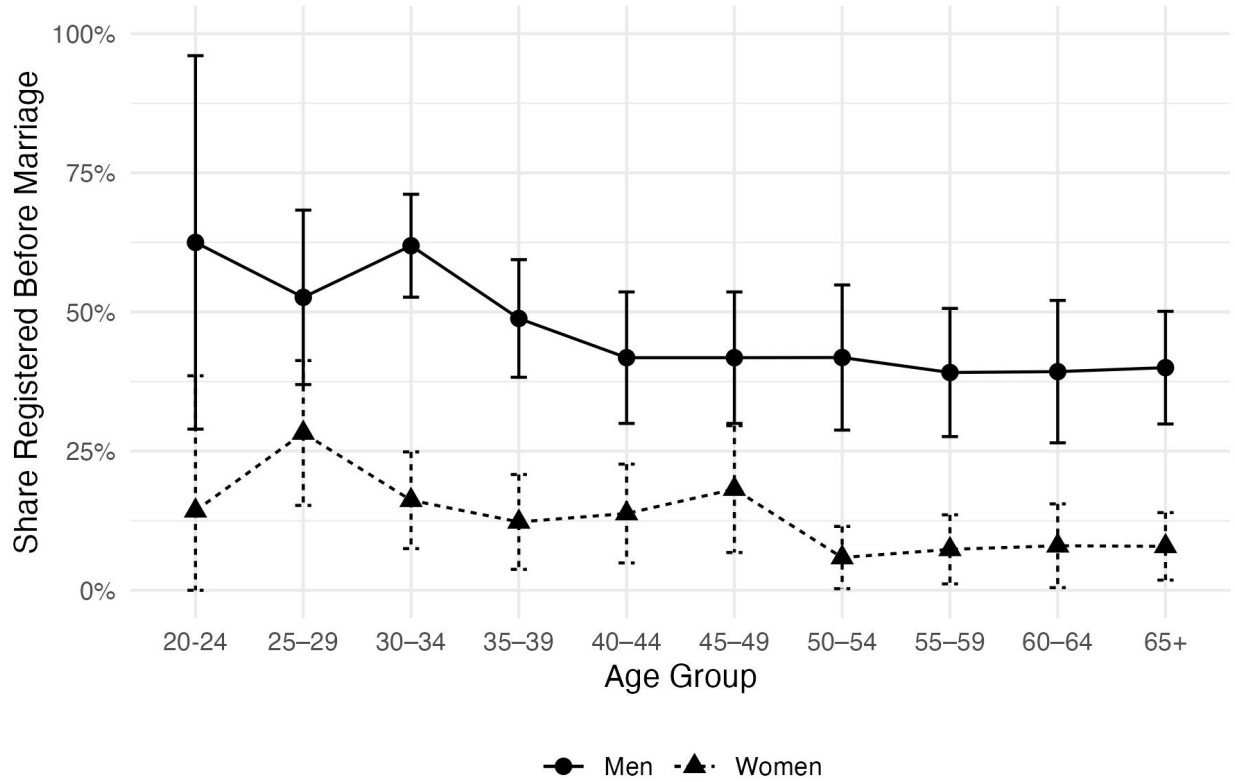
Figure 6: Age Distribution of Registered Voters by Timing of First Registration



The cohort change becomes even more apparent when I look only at ever-married registered voters. Because unmarried registered voters by definition enrolled before marriage, zeroing in on only those who are both married and registered makes for a more targeted comparison. Figure 7 plots the share who first registered *before* marriage among all those who were both married and enrolled at the time of my census. Among married registered women (dashed line), the proportion that first enrolled before marriage is much higher in younger cohorts, peaking at around 28 percent for 25-to-29-year-olds, and drops sharply to around 6 percent for those who were 50 or older at the time of the census. This suggests that over time, women have become more likely to register before marriage. Among men (solid line) the share enrolled before marriage is higher for younger groups as well, with more than 50 percent of those under 40 years registering before marriage compared to only about 40 percent in older cohorts. The figure also suggests that younger cohorts of men and women, particularly those up to 30 years, might converge in their likelihood to register before marriage, although the N of young married men is small in my sample.

What might explain the cohort change in registration timing? My data does not allow me to properly adjudicate this question. But independent of its cause, this observed cohort shift in registration patterns has important implications both for the accuracy and completeness of voter lists. When families deferred women's registration in anticipation of marriage, it produced female under-enrollment. But as registration moves earlier, that same woman is now correctly registered before marriage – but then departs and her registration becomes deadwood at origin. As younger women increasingly enroll in their natal villages before marriage, more entries on these lists are bound to turn obsolete. Unless deletions keep pace with this trend – which the next section argues there is little reason to expect – marriage-migration deadwood therefore should grow on the list over time.

Figure 7: Share First Registered Before Marriage by Age Group and Gender



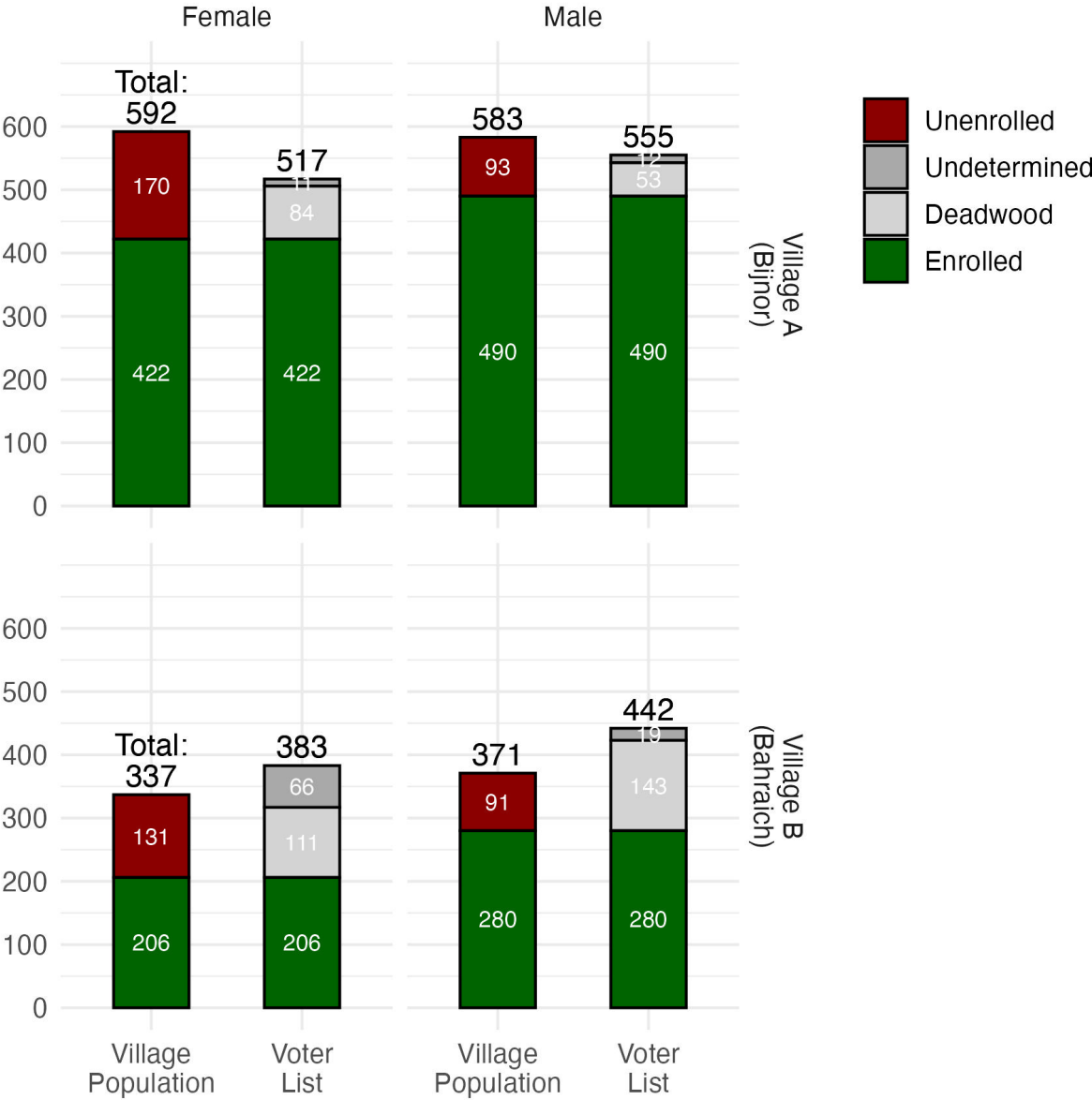
5.5 Why the Aggregate Hides Missing Women

The preceding sections have documented a gendered pattern from two directions: women are disproportionately missing from the roll and female entries make up a disproportionate share of obsolete entries. Taken together, these two facts not simply speak to who is affected, but demonstrate the very measurement problem at the heart of this paper: When a single group bears both errors at once, those errors offset within that group, and are undetectable in the aggregate.

Figure 8 places each village’s resident population beside its voter roll, separately for women and men. In Village A, aggregates suggest modest under-enrollment for both women (12 percent) and men (5 percent). But the record-level match reveals that almost 29 percent of female residents were in fact unregistered, against 16 percent of male residents (red segment) – under-enrollment two to three times higher than aggregate implies. The true extent

of this under-enrollment is masked by offsetting deadwood and undetermined entries on the voter list (grey segments). Of the female entries on the list, 16.2 percent were deadwood and 2 percent undetermined. For male entries, 9.5 percent were deadwood and 2 percent undetermined. The obsolete female entries pad the roll's female count back up, hiding the resident women who are missing from it.

Figure 8: Compositional Difference by Gender and Village



In Village B, aggregate comparisons are even more misleading. Comparing residents to

voter list entries, the roll appears to exceed the population, with 12 percent more female entries than residents and 16 percent for males. Yet almost 39 percent of women and 24.5 percent of men were unregistered in the location (red segment). This under-enrollment is entirely concealed by obsolete entries on the list (grey segments). Among female entries, 29 percent were deadwood and 17 percent undetermined; for male entries, it was 32.4 percent and 4 percent, respectively. Obsolete names not just offset the missing women but *outnumber* them in Village B.

What the two villages illustrate is a general limitation. Disaggregating the aggregate — comparing women on the roll to women in the population, rather than totals to totals — does not recover the gross error. The consequence is institutional, not merely analytical, because the gender balance of the roll is a metric the state itself uses. The Election Commission of India codifies the gender ratio as an indicator of electoral-roll health, computed at the polling-station level against a projected population. In fact, an “abnormal gender ratio” on the list triggers field verification by superiors (Election Commission of India 2023, p. 92). Using 2011 census data, I can compute rough census-based gender ratios and compare them to the gender balance on the voter list.¹³ The ratio of women to men in Village A was .956 according to the 2011 census; the ratio of female to male names on the voter list was .932. In Village B, the census ratio was .881 compared to a the voter list ratio of .867. While the *Manual on Electoral Rolls* does not define which discrepancy, exactly, will trigger a review, a difference of 2 to 3 percentage points, as in my study sites, should be unlikely to raise a flag. But when I look at correctly registered residents — only those registered voters that actually exist — the picture is far more skewed. The ratio of correctly registered female to male adults in Village A is only .861; in Village B, it is .736. Obsolete entries, therefore, do not merely coexist with gendered under-enrollment — they disguise it.

¹³ I am using full population figures, i.e. the number of all females and all males in the location, to calculate the gender ratio. Based on publicly available census data, I have no way of separately calculating the gender ratio for *adults* in each village.

6 Suggested Mechanism: Perverse Bureaucratic Incentives

So far, I have established what the errors in voter registration are and that aggregate monitoring cannot detect them. Here, I provide an explanation for why they arise. My answer centers on the frontline bureaucrats who maintain the electoral rolls and on the incentives they face. This account is not based on data from the two villages, but on interviews with ECI officials, district administrators, and BLOs, as well as the documented structure of the monitoring system. I offer this account as a plausible mechanism consistent with the evidence, though not testable with my data.

One of the most pivotal actors in maintaining India's electoral rolls are Booth-Level Officers (BLOs), who are tasked with conducting regular door-to-door surveys to determine who should be added to or deleted from the list. Yet the role is under-institutionalized for one that is so central. Unlike other officials in the election hierarchy, the BLO position is not established by law, but merely by ECI directive. BLOs can be drawn from a wide range of government and semi-government employees, in urban areas most often government school teachers and in rural areas, *anganwadi* (child health care) workers.¹⁴ They perform their BLO duties on top of their regular employment, and are paid a nominal honorarium of 6,000 rupees a year (≈ 75 USD). Several BLOs I interviewed said they received their remuneration late or only partially, and most considered it too little for the work involved. Many described doing their canvassing at night, on weekends, and on holidays because they work full-time during the day. BLOs therefore hold crucial responsibility as the only field-level election officials, but have minimal institutional backing.

In principle, BLOs are supposed to keep the roll up-to-date by adding those who have become eligible or moved in, and remove those who have died or moved out. In practice, the incentives they face pull away from removal and toward (limited) addition. Four distinct

¹⁴ Based on availability of qualified workers, however, BLOs could also include electricity bill readers, panchayat secretaries, or postmen, among others.

features of the system produce this asymmetry. First, removing a name is difficult. To delete a deceased voter, a BLO needs a death certificate, which is hard to obtain in India. To delete a migrant, they need to either reach the migrant at the new address to verify their move, or have several family members certify the departure – which is difficult if the entire household relocates or if the family holds out hope that the migrant might return. Deletions also carry a risk: if the BLO initiates a deletion and anyone objects – the migrant, a family member, a local politician – the officer might get reprimanded.

Second, additions are comparatively easy. The documentation a new registration requires, such as an ID and proof of residence, are usually relatively easier to obtain than a death certificate. In addition, no one has reason to object to an eligible adult registering correctly.

Third, the targets against which BLOs are judged are set on list totals, not changes to the roll. Supervisors specify the enrollment figures a list should reach, such as the overall count or a gender ratio derived from population projections (Election Commission of India 2023, p. 54, p. 84). But since supervisors do not have reliable, up-to-date data on who many peoples have died or moved, they cannot specify the number of additions and deletions required to get to that count. The one check on changes is perverse: Assistant Electoral Registration Officers are required to audit the 20 polling stations with the highest numbers of additions and deletions (Election Commission of India 2023, p. 92), so a BLO who cleans the list thoroughly draws scrutiny.

Fourth, supervisors rely entirely on aggregate data to evaluate a BLO's work. Superior officers compare population projections to enrollment totals. They do not verify individual names or examine the resident population directly. The source of this incentive structure is precisely the aggregate monitoring instrument described in the previous section. In the aggregate, a roll padded with deadwood that misses large shares of residents and a roll that is genuinely accurate may look indistinguishable to a supervisor.

To illustrate, consider a polling station with 1,000 registered voters on the list. Between revisions, 20 individuals became eligible, 8 moved away and 4 died. The BLO should therefore

delete 12 names and add 20 new ones, for a total of 1,008 entries. However, because deletions are hard, while additions are easy; and because superiors do set targets only for totals – here, 1,008 – and rely on aggregate instruments for quality control, the BLO has incentives to simply add 8 new voters and then stop. This not only leaves 12 deadwood entries on the roll, but also leaves out 12 eligible adults. In fact, deadwood does not merely *hide* under-enrollment, it *produces* it: because BLOs work toward an absolute target, every obsolete name on the list means one less slot to fill with a new addition. One BLO told me that much during an interview: Before a recent state election, he was told by his supervisor that based on population projections, his polling station should have 20 more voters on the list. To him, that meant that after adding 20 names, he could stop surveying the village.¹⁵ The target neither set incentives for the BLO to delete existing entries from the list, nor to conduct a full village canvass to enroll all newly eligible adults.

7 Conclusion

This article uses a record-to-record match of two full village censuses and the official voter lists to reveal what aggregate numbers conceal: both villages suffer from substantial under-enrollment *and* both rolls exhibit substantial deadwood. Importantly, deadwood on the list and under-enrollment in the population not just co-exist, but offset in the aggregate and are therefore invisible to the aggregate benchmark. In Village A, the aggregate would have suggested a modest shortfall in registration, but almost one quarter of residents were unenrolled, partly masked by deadwood on the list. In Village B, where the roll exceeded the population and therefore appeared inflated without suggesting under-enrollment, almost a third of all residents were unregistered, while their absence was masked by still larger deadwood.

My contribution is not the error rate for two villages, which cannot be extrapolated to India. What I demonstrate is that the aggregate benchmark itself – the comparison of the

¹⁵ Phone interview, Uttar Pradesh, March 4, 2022.

eligible population to total enrollment – cannot assess the quality of a voter roll once that list contains deadwood. Because the aggregate only observes the *net* difference between the roll and the population, it is blind to the *gross* errors. In other words, a list can match the size of the population exactly and still be badly composed. This is true even at the relatively fine-grained level of polling-station-wise electoral rolls. It also still holds when aggregating by demographic dimensions along which both errors run, such as gender.

This has direct consequences for how we assess roll quality. A substantial body of work – both in the comparative study of registration quality and the specific literature on Indian voter lists – gauges completeness by setting enrollment totals against population figures (Retnakumar 2009; Shariff and Saifullah 2018; Roy and Sopariwala 2019; Retnakumar and Hari Kurup 2021; James and Garnett 2024). What is more, even the Election Commission of India uses these aggregates to assess the health of its electoral rolls, and to discipline its election officials. Yet any such aggregate comparisons rest on an assumption that there is no deadwood present on the list; if it is – and my data suggest deadwood is far from negligible – any aggregate comparison will systematically underestimate under-enrollment.

This finding also carries a theoretical implication for how under-enrollment is understood. The dominant account, developed largely in the U.S. context, treats non-registration as a problem of cost: individuals fail to enroll because the process requires time, effort, information, and the likes; and some parts of the population will be sensitive even to minor costs (Rosenstone and Wolfinger 1978; Squire et al. 1987; Highton and Wolfinger 1998; Highton 2004; Ansolabehere et al. 2012, see, for example,). In this view, under-enrollment is a matter of demand: who is willing and able to bear the cost of registering. That framework does not theorize deadwood. By contrast, deadwood is a failure of the state to remove obsolete names from the list. Where the state cannot “see” its population accurately, it cannot detect major events – like death or migration – that matter for keeping registers current (Scott 1998; Lee and Zhang 2017; Brambor et al. 2020). Yet I argue that under-enrollment and deadwood are not separable problems. Where the state lacks the information capacity to

keep its voter lists current, deadwood accumulates, and that accumulation can both conceal *and* produce under-enrollment. In such a setting, under-enrollment cannot be explained by registration costs alone; it also depends on the state's capacity to maintain the register. A purely cost-based account becomes inadequate the moment deadwood is present. This is why the coexistence of the two errors matters beyond measurement: under-enrollment is not just a failure of individual voters but at least in part a failure of state capacity.

In my data, both under-enrollment and obsolete entries are gendered. Unlike other demographic findings from my two study sites, this, I argue, is a general pattern produced by the interplay of gendered marriage norms and hyper-local voter registration in India. Patri-local, village-exogamous marriage norms require women to marry outside their natal village and to join their husband's household upon marriage. Because voter lists in India are maintained at the level of the polling station, even minor moves require re-registration. The same institution has the potential to produce both female under-enrollment – if women (or their families) hold off on registering until after the marital move – *and* female deadwood – if women do enroll before marriage, their entries necessarily turn into deadwood after their wedding. Because female omissions and female deadwood offset within the female electorate, the roll can reproduce the population's size *and* gender balance almost exactly while still being badly composed.

The reason why deadwood persists on Indian voter lists, I suggest, are the perverse incentives that the frontline bureaucrats in charge of maintaining the voter lists face. For these Booth-Level Officers, removing a name from the list is difficult and risky, while adding names is easy. Because their work is judged against aggregate totals, BLOs have reason to let deadwood accumulate and to enroll only enough new voters to meet their targets. The same aggregate monitoring that fails to detect the roll's errors therefore also shapes the behavior that creates them. This mechanism rests on interviews and official election commission documents rather than on my data. Testing it systematically would require data on how BLOs perceive and respond to incentives.

My findings carry practical implications that follow directly from the measurement argument. If aggregate comparisons cannot detect the errors on a voter list, then improving roll quality requires micro-level checks. These checks need not be as costly as a full census. As part of my study, I also asked each village’s *pradhan* (village council president) to annotate the voter lists. Village A’s *pradhan* was able to identify 90 percent of deceased individuals; Village B’s identified roughly half of all deaths, almost three-quarters of all marriage migrants, and 12 percent of other migrants with minimal effort. This is only suggestive, of course, and local officials may have different levels of knowledge about their local populations (Schneider 2019). But it indicates a path to cheap, ground-level verification that aggregate monitoring cannot deliver.

This study comes with several limitations. Two villages cannot generate representative estimates of roll quality in India, and I claim no generalizability beyond the gender finding. Nor do I make claims on how common these errors are. Instead, what my two villages demonstrate is proof of concept: in randomly selected polling stations, aggregate benchmarks *can* hide large offsetting errors; and aggregate checks have no way of recovering them.

Finally, my findings raise questions about another aggregate measure. India’s high and rising voter turnout is often cited as evidence of the country’s democratic vitality. But turnout in India is calculated as the number of ballots cast against the number of names on the electoral roll. Turnout rates constructed that way reflect not just how many registered voters turn out (driving up the numerator) but also how much deadwood the list contains (inflating the denominator); and they are entirely blind to the disenfranchisement that faulty voter lists hide. This becomes apparent when I recalculate *actual* turnout in my study villages, by dividing the number of ballots cast in the 2022 election by the number of eligible residents my census enumerated. In Village A, official turnout stood at 56.25 percent, but given the location’s high under-enrollment compared to its share of deadwood, actual turnout among the eligible population was only 51.32 percent. In Village B, where deadwood far outpaced under-enrollment, official turnout was 64 percent compared to an actual turnout of

74.6 percent. Again, the point is not the specific figures, but more general: because turnout is built from the same aggregates, it inherits their unreliability. What I show is not that, or by how much, official turnout statistics are off, but that they *can* be wrong – in either direction.

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Online Appendix

A Residency Rules in Indian Voter Registration

I classified all those who a) were identified as migrants by family members, b) were not claimed as ordinarily resident by any interviewed household, c) did not maintain their own homestead in the village, but were d) *not* marriage migrants as “other migrants.” My classification criteria conform with the legal requirement to maintain an “ordinary residence” in a location to register to vote. Yet there is considerable debate about what exactly that means. The ECI’s own *The Manual on Electoral Rolls* acknowledges that “the concept of ‘ordinary residence’ has not been defined clearly anywhere in the statutes” (Election Commission of India 2023, p. 42). Following several court verdicts on the question, the ECI requires that a person should be usually staying in a place and intent to remain there for a considerable time; temporary absence, however, does not disqualify (p. 42). While that means that seasonal and cyclical migrants could have a reasonable claim to be registered in their place of origin, any permanent and potentially even semi-permanent¹⁶ migrants would not.

But given how difficult many economic migrants find it to enroll as voters at their destinations, many may intentionally choose to keep their registration at their place of origin to have at least *some* option for political participation (Kumar and Banerjee 2017; Thachil 2017; Gaikwad and Nellis 2021b,a). In fact, Gaikwad and Nellis (2021b) showed that the vast majority (98 percent) of eligible urban migrants indicated they would like to be registered to vote at their current residence –independent of feelings of attachment to their home region – but that bureaucratic hurdles prevented them from doing so. Remaining on the voter list in their place of origin, then, is a fallback option to prevent full political exclusion rather than a purposeful choice of where to exercise their vote. That is particularly true for precarious labor migrants, many of whom are unable to take time off and forgo wages during elections to travel to their place of origin and cast a ballot.¹⁷ Accordingly, election officials might be reluctant to remove them from the list.

Yet many economic migrants keep their origin registration because they find it difficult to enroll at their destination (Kumar and Banerjee 2017; Thachil 2017; Gaikwad and Nellis 2021b,a). Removing them from the list at origin potentially prevents these migrants from political participation all together, which is why the ECI might be reluctant to do so. While I can be just as certain about the classification of “other migrants” as about the other types of deadwood, this category then is more contestable. This essentially highlights the tension between completeness and accuracy of the voter list. While I can be just as certain about the classification of “other migrants” as about the other types of deadwood, this category then is more contestable.

¹⁶ Tumbe (2018) defines semi-permanent migrants as those who move for economic reasons, stay at their destination most of the year for many consecutive years, but intent to eventually return to their place of origin to retire. Most of these semi-permanent migrants are able-bodied men, although they sometimes move with their wife and children as well.

¹⁷ See, for example, *Press Information Bureau, Government of India*, December 29, 2022, “[ECI Ready to Pilot Remote Voting for Domestic Migrants; Migrant Voter Need Not Travel Back to Home State to Vote](#)”; *CNN*, May 13, 2024, “[In the World’s Biggest Election, Millions of Migrants Are Unable to Vote](#).”

Tables

Table A.1: Descriptive Statistics for the Two Study Sites

	Village A (Bijnor)	Village B (Bahraich)
Panel A. Resident Adult Population (Census)		
N Adults (18+)	1175	708
% Female	50.38	47.60
Avg Age	40.37	41.26
Avg Years Schooling	5.54	3.91
% Hindu	5.11	93.93
% Muslim	94.89	6.07
% General	0.00	28.81
% OBC	94.89	53.39
% SC	5.11	17.80
Avg Monthly HH Income (INR)	11377	6345
Avg N Adults/HH	3.68	2.96
Panel B. Voter List Entries (Electoral Roll Annotation)		
N Entries on Roll	1072	825
% Female Entries	48.13	46.42
Avg Age of Entries	42.76	45.32

Notes: Panel A describes the enumerated resident population; Panel B describes the entries from the electoral roll. Household income is approximate and self-reported.

Table A.2: Characteristics of Voter List Entries by Category

	Village A (Bijnor)	Village B (Bahraich)
Panel A. Deadwood		
Clerical Error		
% Female	37.5%	38.1%
Avg Age	33.5	32.0
Death		
% Female	57.5%	63.5%
Avg Age	79.4	68.6
Marriage Migration		
% Female	100.0%	100.0%
Avg Age	26.4	29.5
Other Migration		
% Female	26.2%	29.7%
Avg Age	34.4	43.6
Panel B. Undetermined Entries		
Undetermined		
% Female	47.8%	77.6%
Avg Age	33.3	41.4

Notes: The table reports the percentage of entries labeled as female and the average age reported for each entry on the official voter list.

Table A.3: Correlates of Enrollment, by Village

	Village A Only	Village B Only
Female	−0.109*** (0.024)	−0.126*** (0.031)
Age (years)	0.013*** (0.001)	0.009*** (0.001)
Education (years)	0.010*** (0.003)	0.000 (0.006)
HH income (1,000 INR)	0.001 (0.002)	0.001 (0.002)
OBC		−0.075 (0.064)
SC		−0.058 (0.066)
Num.Obs.	1175	708
R2	0.185	0.151

Robust standard errors clustered by household in parentheses. Religion and caste are excluded from Village A because they do not vary within village; religion is excluded from Village B for the same reason. Reference category for OBC and SC is general caste group. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Figures

Figure A.1: Study Sites for Village Censuses and Voter Survey

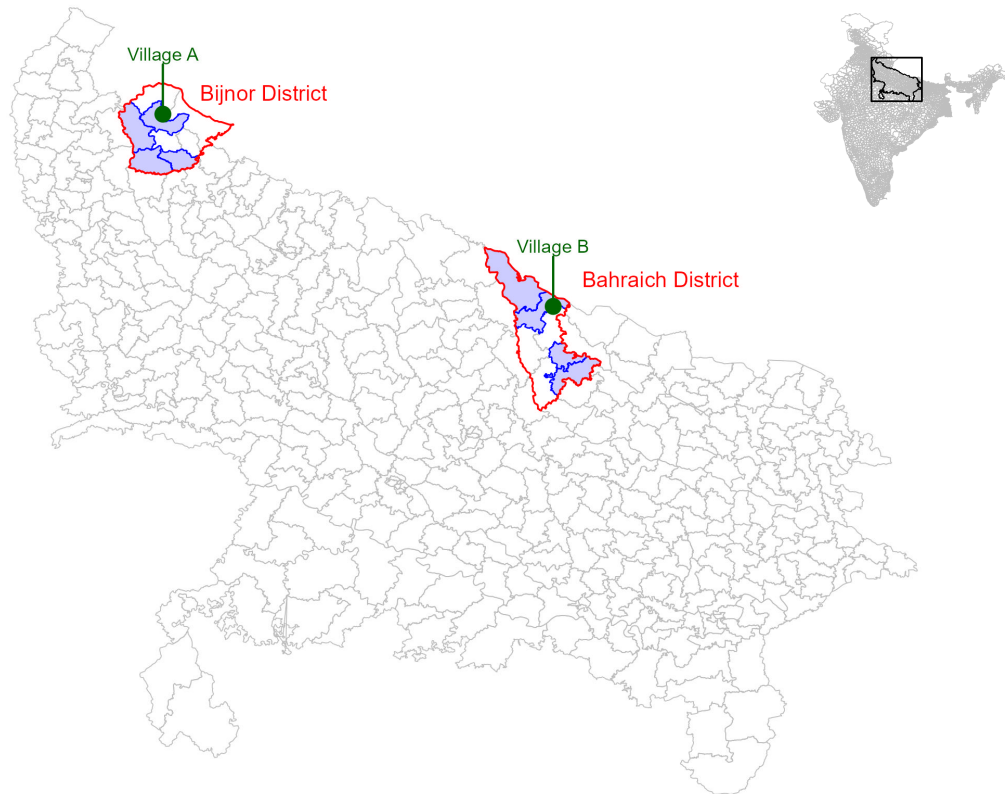
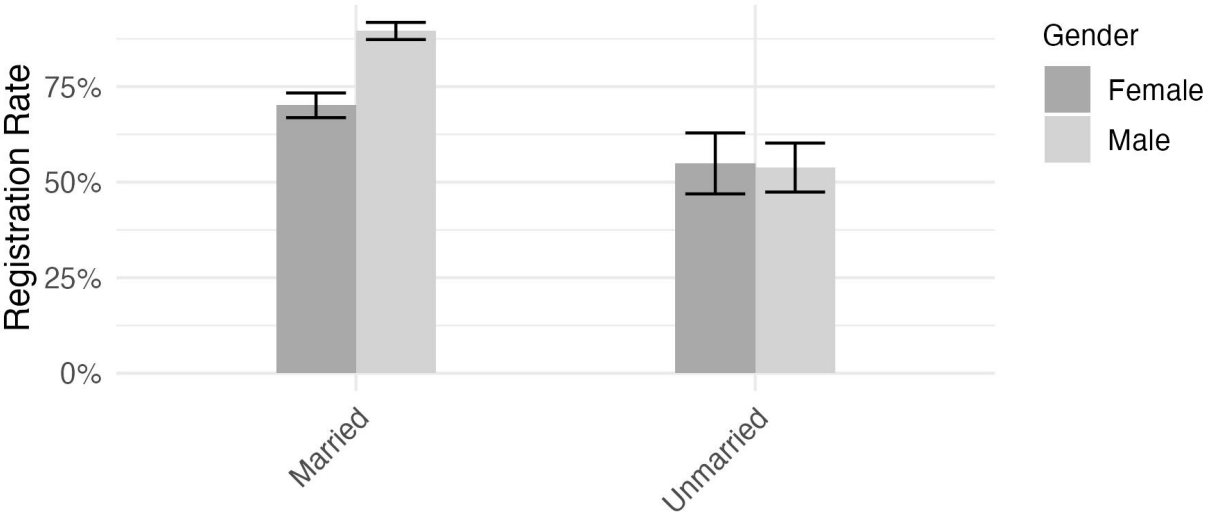


Figure A.2: Registration Rate by Marital Status and Gender



Error bars show the 95% confidence interval.